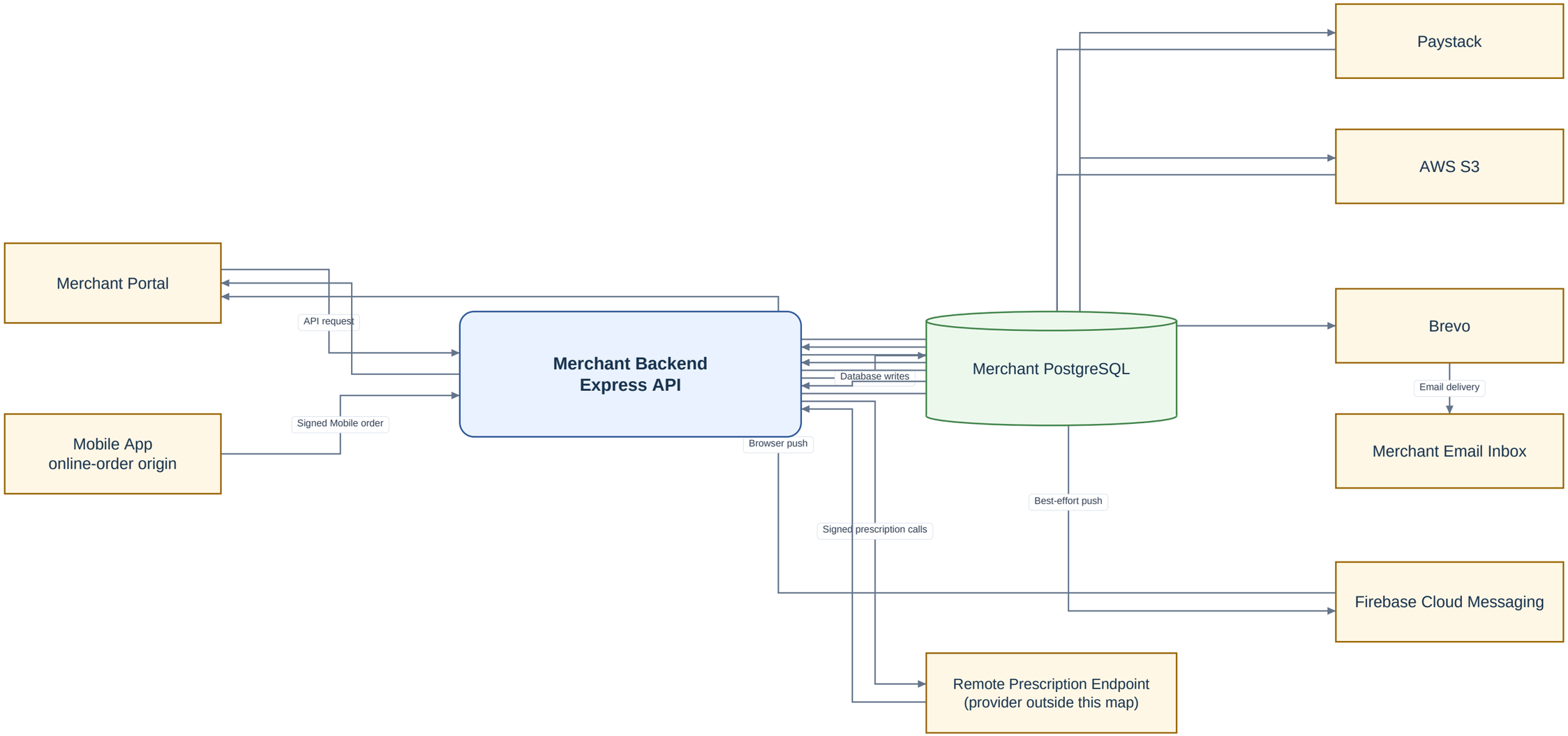
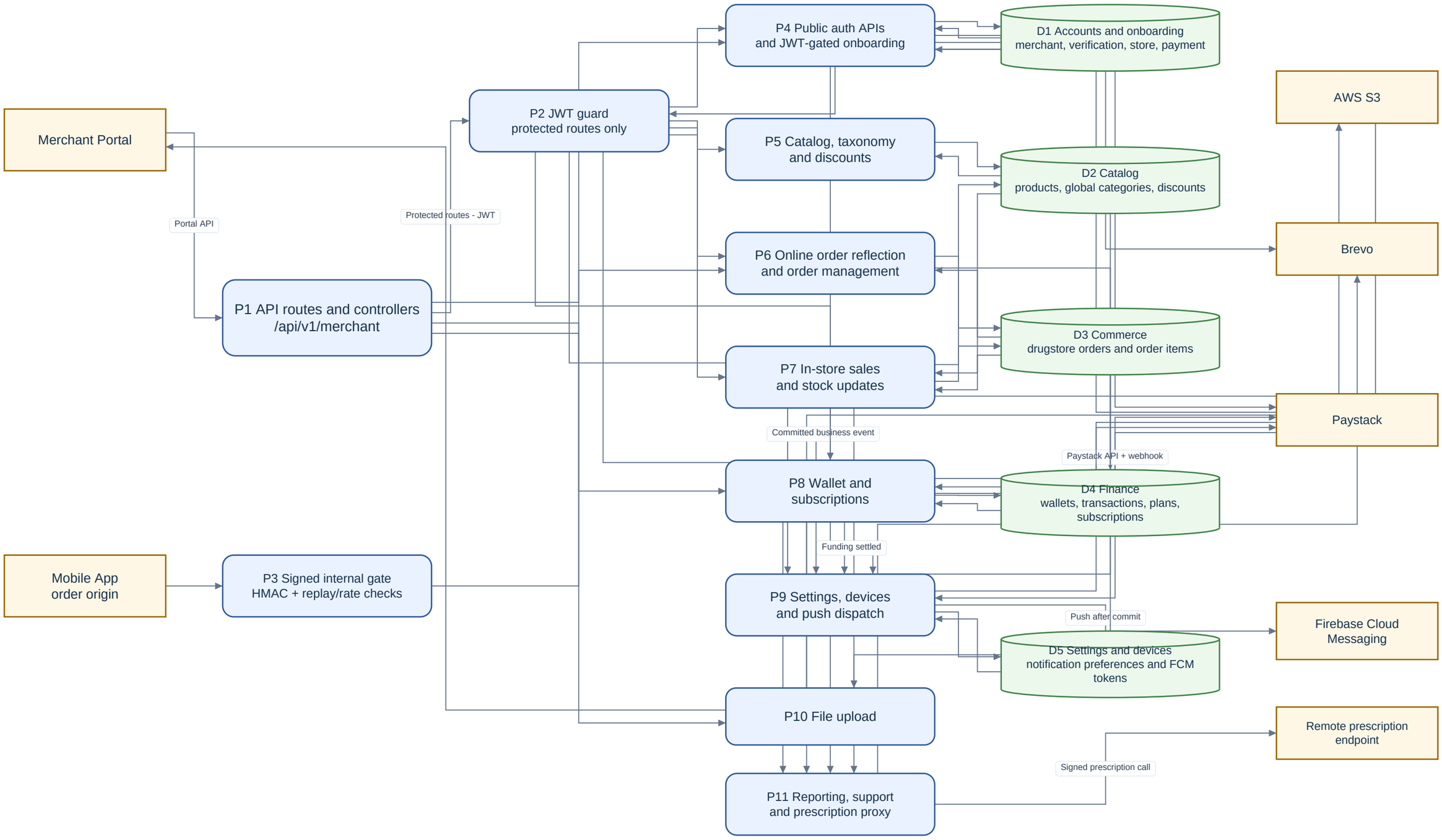
Context - Level 0



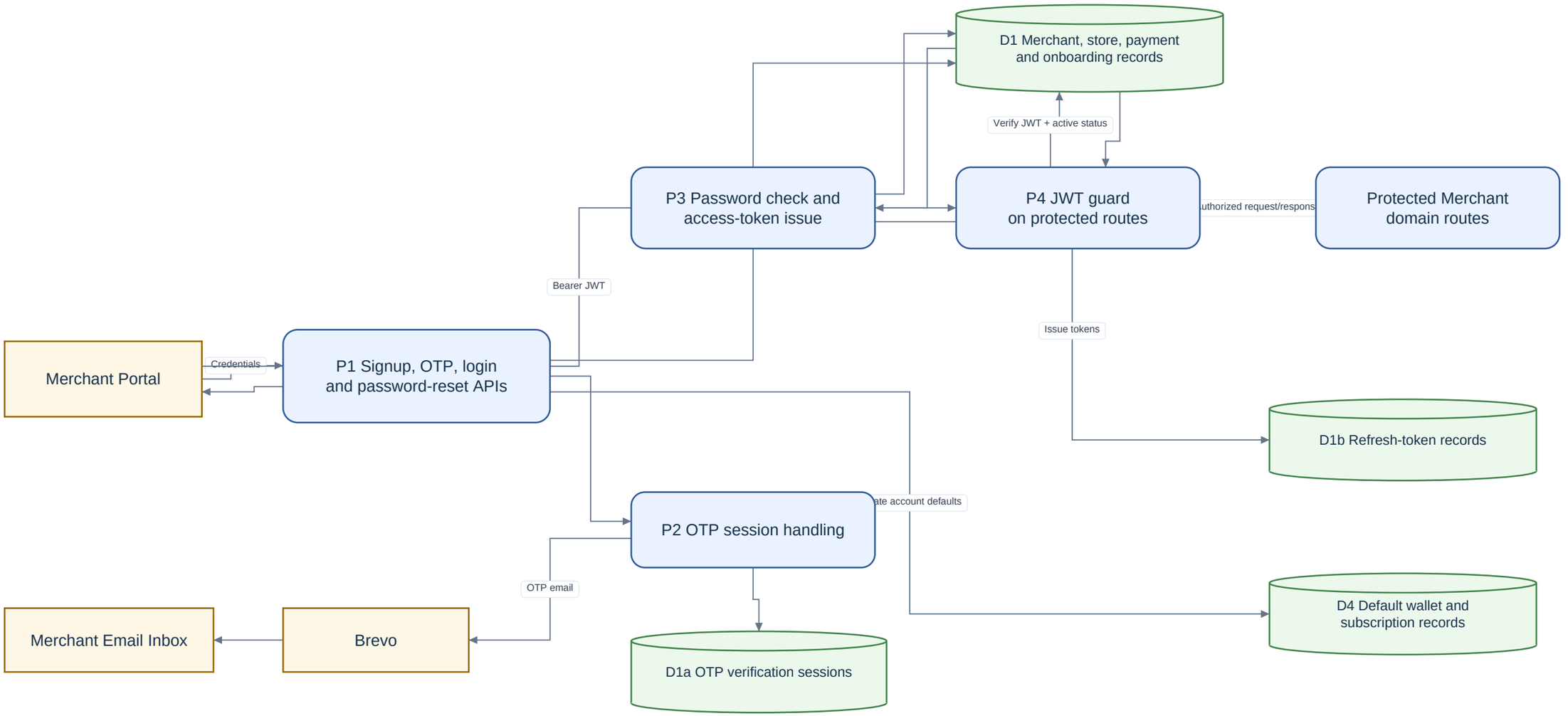
Scope note: Mobile App is the order's business origin. Merchant receives an internally signed integration request; the upstream forwarding mechanism is not decomposed here.

Merchant Overview - Level 1



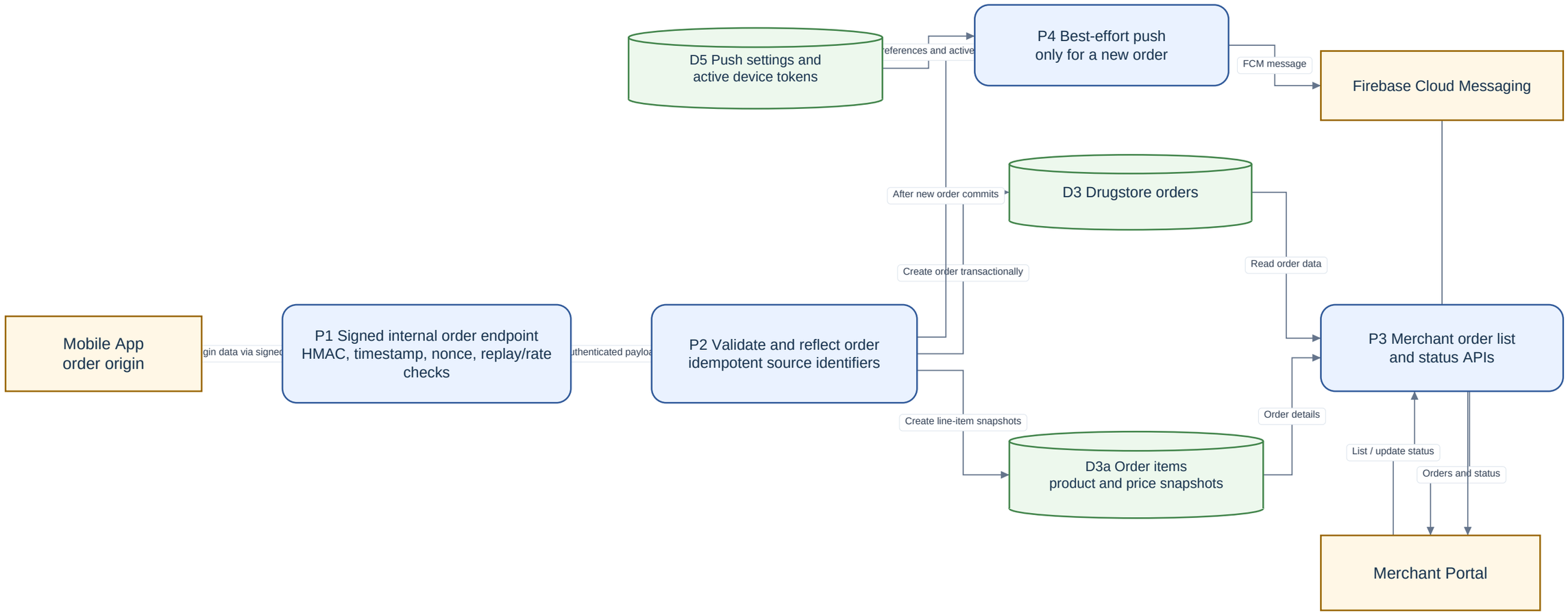
Access is route-specific: auth APIs and uploads have their own behavior; protected routes use JWT; internal order ingress uses HMAC and rate limiting; Paystack webhooks verify signatures. Taxonomy JSON syncs into D2 at startup.

Authentication and Portal Access



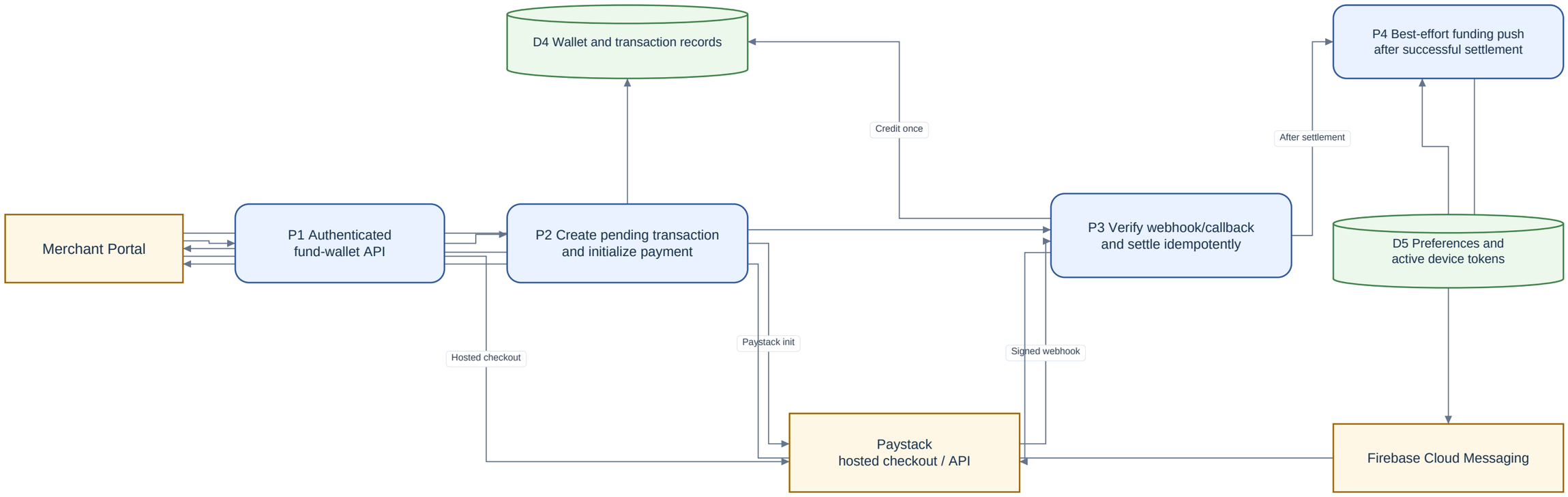
Signup stores/emails an OTP first; verified completion creates the Merchant and defaults. Login returns an access token and a persisted refresh token. The JWT guard verifies the token and checks the Merchant is active.

Online Orders from Mobile App



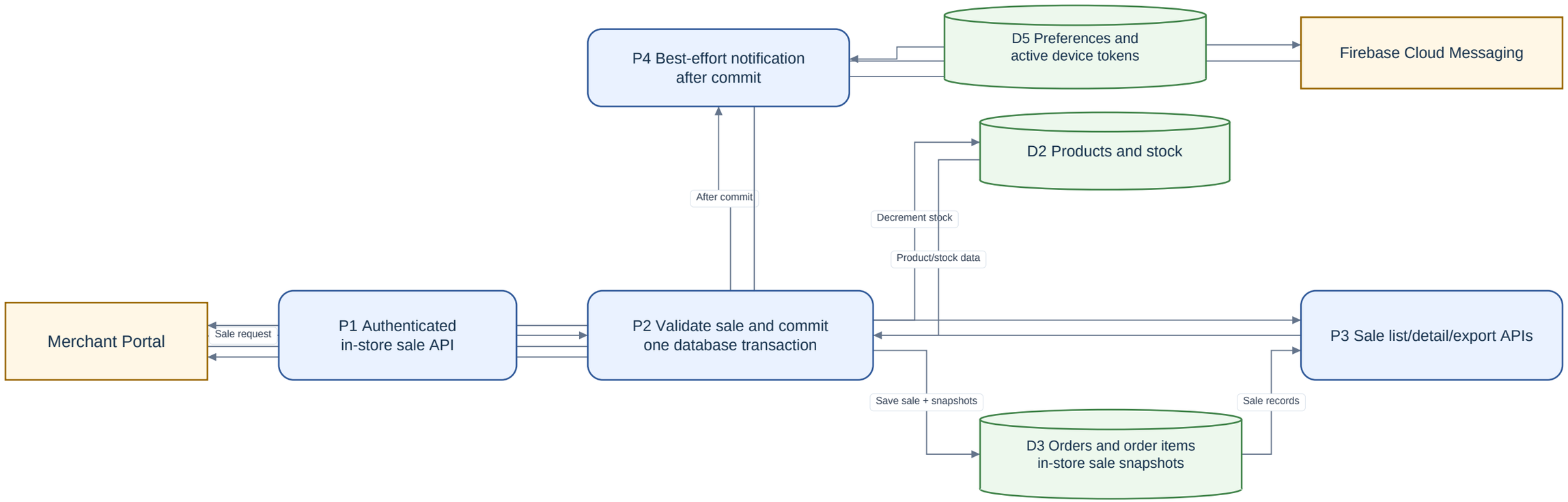
Mobile App identifies the online order's business origin. The upstream forwarding path is outside this map; the Merchant endpoint itself requires a signed request. Merchant stores supplied payment fields and snapshots but does not charge the buyer in this flow.

Wallet Funding



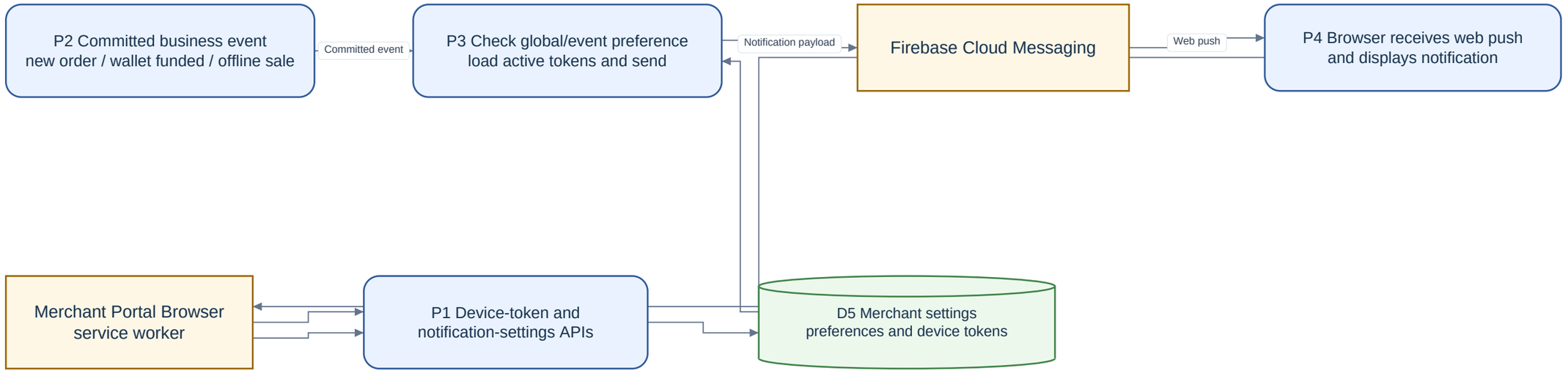
Paystack webhook signatures are verified over the captured raw body. The authenticated confirm-funding path verifies the reference with Paystack. Both converge on one idempotent settlement; FCM failure does not undo a successful credit.

Offline Sale



The service checks inventory and writes stock changes, the sale order, and line-item snapshots in one transaction. Push is attempted after commit; a failed push does not roll back the sale.

Push Notification Setup and Delivery



Business data is committed before notification delivery is attempted. Delivery can be skipped by preferences, no active devices, or Firebase configuration, and failures do not undo the order, sale, or wallet credit.